

# FAILSAFE

AI-Powered Student Failure Risk Prediction System

|        |         |      |         |          |            |
|--------|---------|------|---------|----------|------------|
| Python | XGBoost | SHAP | FastAPI | React.js | PostgreSQL |
|--------|---------|------|---------|----------|------------|

|                                       |                             |                  |                  |
|---------------------------------------|-----------------------------|------------------|------------------|
| Dataset<br>UCI Student<br>Performance | Model<br>XGBoost Classifier | Accuracy<br>~90% | ROC-AUC<br>~0.94 |
|---------------------------------------|-----------------------------|------------------|------------------|

ABSTRACT

FAILSAFE is an end-to-end AI-powered web application that predicts student academic failure risk early in the semester using machine learning and Explainable AI. Built on XGBoost and SHAP, it flags at-risk students before final examinations, explains the reasons behind every prediction in plain language, and auto-generates personalised intervention plans — enabling faculty to act before it is too late.

PROBLEM STATEMENT

Traditional academic monitoring detects failure only after final grades are published, leaving no time for corrective action. Faculty lack proactive, data-driven tools to identify struggling students mid-semester, understand the root causes of their difficulties, or deliver targeted support.

FAILSAFE solves three specific problems:

1. **Predict** failure risk early using attendance, grades, and behavioural data.

2. **Explain** every prediction using SHAP so faculty understand and trust the results.

3. **Act** by auto-generating personalised intervention plans per student.

DEVELOPMENT APPROACH

The system is built across four integrated layers:

- ML Pipeline** — Data loading, preprocessing with scikit-learn, XGBoost training, SHAP computation, artifact persistence.
- Backend API** — FastAPI with JWT auth, SQLAlchemy ORM, PostgreSQL, role-based access for Faculty and HOD.
- Database** — PostgreSQL with soft-delete (is\_deleted) across all tables, settings storage, prediction history.
- Frontend** — React.js dashboard with Recharts visualisations, SHAP bar charts, intervention tracking, dark/light mode.

DATASET

UCI Student Performance Dataset (Kaggle) — **student-mat.csv**, 395 records, 33 features covering academic, family, lifestyle, and behavioural attributes. Target: G3 < 10 = Fail (binary classification).

|                                               |                                                     |                                                      |                                                 |                                                   |                                                   |
|-----------------------------------------------|-----------------------------------------------------|------------------------------------------------------|-------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| <div><div>~90%</div><div>Accuracy</div></div> | <div><div>~0.94</div><div>ROC-AUC Score</div></div> | <div><div>316</div><div>Training Samples</div></div> | <div><div>79</div><div>Test Samples</div></div> | <div><div>33</div><div>Input Features</div></div> | <div><div>200</div><div>XGBoost Trees</div></div> |
|-----------------------------------------------|-----------------------------------------------------|------------------------------------------------------|-------------------------------------------------|---------------------------------------------------|---------------------------------------------------|

XGBOOST CONFIGURATION

XGBoost (Extreme Gradient Boosting) builds sequential decision trees where each tree corrects errors of the previous one. It was selected for its excellence on tabular data, native SHAP support, and class imbalance handling.

| Parameter        | Value   | Purpose                   |
|------------------|---------|---------------------------|
| n_estimators     | 200     | Number of boosting trees  |
| max_depth        | 6       | Maximum tree depth        |
| learning_rate    | 0.05    | Shrinkage factor          |
| subsample        | 0.8     | Row sampling per tree     |
| colsample_bytree | 0.8     | Feature sampling per tree |
| scale_pos_weight | auto    | Handles class imbalance   |
| eval_metric      | logloss | Training objective        |

PREPROCESSING PIPELINE

All preprocessing artifacts are saved alongside the model and applied identically during real-time inference.

|                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Categorical Encoding</div><div>LabelEncoder applied to 16 categorical features (gender, jobs, yes/no fields). E</div></div> |
| <div><div>Numerical Scaling</div><div>StandardScaler normalises 15 numeric features to mean=0, std=1. Scaler saved</div></div>        |
| <div><div>Target Creation</div><div>G3 &lt; 10 → label 1 (Fail). G3 &gt;= 10 → label 0 (Pass).</div></div>                            |
| <div><div>Class Balancing</div><div>scale_pos_weight = negative count / positive count. Prevents model bias toward</div></div>        |
| <div><div>Train/Test Split</div><div>80/20 stratified split preserving class ratio in both sets.</div></div>                          |
| <div><div>Feature Removal</div><div>G3 excluded from features to avoid data leakage. Only G1 and G2 used as gra</div></div>           |

RISK SCORE INTERPRETATION

| Risk Level  | Score Range | Meaning                                | Action                                |
|-------------|-------------|----------------------------------------|---------------------------------------|
| HIGH RISK   | ≥ 65%       | Student very likely to fail final exam | Immediate counselling + interventions |
| MEDIUM RISK | 35–65%      | Elevated risk, needs monitoring        | Support measures + weekly tracking    |
| LOW RISK    | < 35%       | Performing within acceptable range     | Continue monitoring each semester     |

SHAP EXPLAINABILITY INSIGHTS & KEY FEATURES

**What is SHAP?** SHAP (SHapley Additive exPlanations) uses game theory to fairly attribute a model prediction across all input features. Each feature receives a SHAP value showing how much it pushed the risk score up (positive) or down (negative) for a specific student. This makes every prediction transparent, auditable, and actionable for non-technical faculty.

TOP SHAP RISK FACTORS IDENTIFIED FROM UCI DATASET

|                                                                                                                                        |                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>+0.31</div><div>0</div></div> <div><b>FAILURES</b><br/>Past academic failures are the single strongest predictor. Each</div> | <div><div>+0.12</div><div>0</div></div> <div><b>STUDYTIME</b><br/>Students studying less than 2 hrs/week show elevated risk. Insufficient</div> |
| <div><div>+0.24</div><div>0</div></div> <div><b>ABSENCES</b><br/>Students missing more than 15 classes show markedly elevated</div>    | <div><div>+0.09</div><div>5</div></div> <div><b>HIGHER</b><br/>Lack of aspiration for higher education reduces motivation. Students a</div>     |
| <div><div>+0.21</div><div>0</div></div> <div><b>G2 GRADE</b><br/>Second period grade is a leading indicator. Scores below 8/20</div>   | <div><div>-0.06</div><div>5</div></div> <div><b>FAMREL</b><br/>Strong family relationships provide emotional and academic support, a</div>      |
| <div><div>+0.18</div><div>0</div></div> <div><b>G1 GRADE</b><br/>First period grade provides the earliest measurable signal of a</div> | <div><div>-0.04</div><div>5</div></div> <div><b>INTERNET</b><br/>Home internet access enables self-directed study and resource access</div>     |

HOW SHAP DRIVES PERSONALISED INTERVENTIONS

| SHAP Factor Detected     | Auto-Generated Intervention                                | Type             |
|--------------------------|------------------------------------------------------------|------------------|
| High failures score      | Structured remedial study plan with weekly milestones      | Study Plan       |
| High absences score      | Daily attendance monitoring + family notification protocol | Attendance       |
| Low G1 or G2 grade       | Peer mentor assignment + practice exam papers              | Academic Support |
| High alcohol consumption | Wellness check + lifestyle counselling referral            | Wellness         |
| No higher edu aspiration | Career counselling + alumni mentorship connection          | Motivation       |
| High risk overall (≥65%) | Immediate one-on-one counselling session with faculty      | Counselling      |

KEY FEATURES IMPLEMENTED

|                                  |                                                                                                        |                                      |                                                                                                   |
|----------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------|
| <div>Early Risk Prediction</div> | XGBoost predicts failure probability from G1, G2, absences, and behavioural data mid-semester.         | <div>SHAP Explainability</div>       | Per-student SHAP values stored in DB and visualised as colour-coded bar charts in the UI.         |
| <div>AI Intervention Plans</div> | Rule engine maps SHAP factors to counselling, study plans, attendance monitoring, and wellness checks. | <div>Bulk CSV Upload</div>           | Faculty upload UCI-format CSV files to import and auto-predict entire cohorts at once.            |
| <div>JWT Authentication</div>    | Secure role-based access with 24-hour tokens. Faculty see own students; HOD sees all.                  | <div>Soft Delete</div>               | is_deleted column on all tables ensures data is archived not destroyed, maintaining audit trails. |
| <div>Dark / Light Mode</div>     | Theme preference stored in PostgreSQL settings table and restored on every login.                      | <div>Edit Student + Re-predict</div> | After editing student data the model automatically re-runs to reflect updated information.        |

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**Conclusion:** FAILSAFE demonstrates how machine learning and explainable AI can transform academic monitoring from a reactive end-of-semester activity into a proactive, data-driven support system. By combining XGBoost's predictive accuracy with SHAP's transparency and an automated intervention engine, the system gives faculty not just a warning but a complete understanding of each at-risk student's situation — and a concrete, personalised plan of action to act before it is too late.